DO NOW — RECALL

Scope & parameter passing in C#

Name: _____ Class: _____ Date: _____

1. What is the difference between a C# function and a procedure-style method?

2. Which C# loop always executes its body at least once?

3. What does == do in a C# condition?

4. In Triple(7) , identify the argument. In Triple(int number) , identify the parameter.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A function returns a value . A procedure-style method uses void and does not return a value.

2. Model answer

do...while Its condition is checked after the body has run.

3. Model answer

It compares two values for equality . = assigns; == compares.

4. Model answer

Argument = 7 · parameter = number The argument is the actual value supplied; the parameter is the named input in the method definition.

By the end, you can…

• explain the scope of local and global/class-level variables;

• explain what happens when a parameter is passed by value;

• use C# ref to pass a variable by reference.

Vocabulary — know these terms

Recall the term — Scope

Recall the term — Local variable

Recall the term — Global variable

Recall the term — Parameter

Recall the term — Argument

Recall the term — Pass by value

Recall the term — Pass by reference

Recall the term — ref

NEW LEARNING

Read, annotate and explain

Scope & parameter passing

```
public class Program
{
public static int score = 0;

public static void Main()
{
int attempts = 3;
}

public static void Change(ref int number)
{
number = 100;
}
}
```

OCR idea: global variables can be accessed across the program; local variables exist only within the subroutine/block in which they are declared.

Scope controls where a variable can be used

A local variable is declared inside a method or block and can only be used within that scope.

For this course, we use a class-level static field to demonstrate the OCR idea of a global variable that can be accessed by multiple methods.

Parameters are local to the method

A parameter receives data when a method is called and behaves as a local variable inside that method.

number exists within Triple . The value supplied in the call is the argument.

By value = the method receives a copy

Changing number changes only the method's local copy. The original variable in Main() is unchanged.

ref gives access to the original variable

Write ref in both places: in the method parameter and in the method call.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Why doesn't the original value change?

Start in Main

```
int score = 10;
```

The variable score belongs to Main() and currently stores 10.

The method receives a copy

```
public static void ChangeByValue(int number)
{
    number = 100;
}
```

The parameter number is a separate local variable initialised with the value 10.

Only number becomes 100

```
ChangeByValue(score);

inside method:
number = 100

back in Main:
score = 10
```

When the method ends, its local parameter is no longer needed.

Reference passing changes the original

```
ChangeByReference(ref score);

// method
public static void ChangeByReference(ref int number)
{
    number = 100;
}
```


WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Classify each variable

Guided 2 — CHECK

Track the two variables
number is a local copy, so changing it does not change x .

Guided 3 — CHECK

Where must ref appear?
Using ref in both places lets the method change the original variable.

Guided 4 — CHECK

Complete both explanations

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Passing an integer parameter by value means changing the parameter does not change the original integer variable.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY: Identify the scope

Which variable can be accessed by multiple methods? Which is local to AddOne() ?

Method / working:

Guided 2 — TRY: Predict pass-by-value

What is output?

Method / working:

Guided 3 — TRY: Make the change persist

Modify the previous example so x becomes 10 after the call.

Method / working:

Guided 4 — TRY: Explain by value vs by reference

OCR expects more than the words “copy” and “address”. Explain what happens to the original data .

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Local variables in two methods
Create two procedure-style methods: First() and Second() .
Inside each method declare a local variable called result , but give it a different value. Output each value.

Question 2

Prove pass-by-value
Set int score = 20; in Main() .
Create SetToZero(int number) that changes number to 0.
Call the method, then output score .

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Change the original with ref
Adapt Challenge 2 so the original score becomes 0.
Rename the method SetToZeroByReference and use ref correctly in both required places.

Question 4

Swap two values
Create Swap(ref int first, ref int second) .
Use one local temporary variable so that the two original values exchange places.
In Main() , start with a = 10 and b = 25 , call the method and output both values.